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Task 03: Final Project Report

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Chapter 1: Introduction and Project Scope

1.1 Introduction

The report at hand provides a joint examination of leakage-controlled malware-family classification via the MaleVis RGB byte-image dataset. It started with a data-quality audit of 14,226 images across 26 categories. This audit reported 187 groups of exact-duplicate images (480 images total), with exact match in the training split: 173 validation images. In order to avoid optimistic evaluation, hash grouped partitions were created with no exact-image overlap and only the 25 malware families from the particular class of samples were modeled. A baseline four-block convolutional neural network (CNN) was set up, and a Convolutional Block Attention Module (CBAM) was included as lightweight channel-and-spatial attention.

The baseline CNN produced 77.25% held-out accuracy and 77.30% macro-F1. For instance, one matched CNN+CBAM model increased accuracy to 79.81% and weighted-F1 to 79.36%, while macro-F1 decreased slightly to 76.77%; this indicated that aggregate gains were not uniform across classes. A four-block CBAM model with batch normalization, dropout 0.40, and geometric augmentation disabled was chosen from subsequent validation-only ablation and tuning. The final model had an held-out accuracy of 92.69%, macro-F1 of 93.00%, weighted-F1 of 92.76% and macro one-vs-rest ROC-AUC of 0.9954. In grouped five-fold cross-validation, the chosen model achieved $96.28\% \pm 0.57\%$ macro-F1 against $93.05\% \pm 2.02\%$ for a matched no-attention model. Though attention swept all five folds, the paired two-sided Wilcoxon result ($W = 0$, $p = 0.0625$) did not cross to $\alpha = 0.05$ line. One correct and one incorrectly predicted image were examined with Grad-CAM and LIME. Thus, the attention effect found is practically but not statistically significant, and its larger final improvement over the full model-selection pipeline.

1.2 Background

Byte-level structures are repeated among malware families. Thus when the pixels of an image are actually mapped to executable bytes, these structures show up as bands, blocks, repetitive textures and noise-like regions. From these representations, a CNN can learn local patterns or an attention mechanism reweighs informative channels and spatial positions. MaleVis facilitates that because it has 25 malware families and one legitimate category, each characterized as an RGB image.

1.3 Objectives

- Check for imbalance, corruption, duplication and split leakage in the dataset;
- Construct a reproducibly custom CNN baseline and matching CBAM extension;
- Choose a last set of parameters using validation rather than the test data used to assess the model;
- Aggregate class-balanced performance, fold variability, and paired statistical proof; and
- Use representative correct and incorrect predictions to interpret image saliency not as causal malware semantics.

1.4 Project Development Record

Task	What was done	Results
Task 1	Exploratory data analysis	Analyze the dataset and found the total images, duplicate groups etc.
Task 2	Baseline CNN and first controlled CBAM comparison	Used these results later for the improvement experiments.
Task 3	Ablation, tuning, grouped CV, final test, Grad and Lime	Selected the final model and compared it with the baseline CNN and CNN+CBAM.

Table 1: Compare Of Tasks

Chapter 2: Dataset and Data Integrity

2.1 Original MaleVis Data

The original dataset consists of 14226 readable RGB PNG images (300×300 pixels): 9100 supplied training images across all categories and 5126 supplied validation images from a subset of the categories [1], [6], [7]. Training part is balanced at 350 images per class. This validation split is imbalanced as for the Other class, which are legitimate, there are 1,482 instances while most of the rest contain just 120–151. Other has 1,832 images and Agent 470 across the entire dataset for a majority-to-minority ratio of 3.90 :1.

Property	Value	Implication
Images / categories	14,226 / 26	Adequate scale for custom CNN experiments
Image format	300 × 300 RGB PNG	Uniform geometry; retain RGB channels
Corrupted / missing	0 / 0	No file-repair stage required
Most / least populated	Other 1,832 / Agent 470	Accuracy alone is insufficient
Original split	9,100 / 5,126	Validation is dominated by Other

Table 2: MaleVis Data Analysis

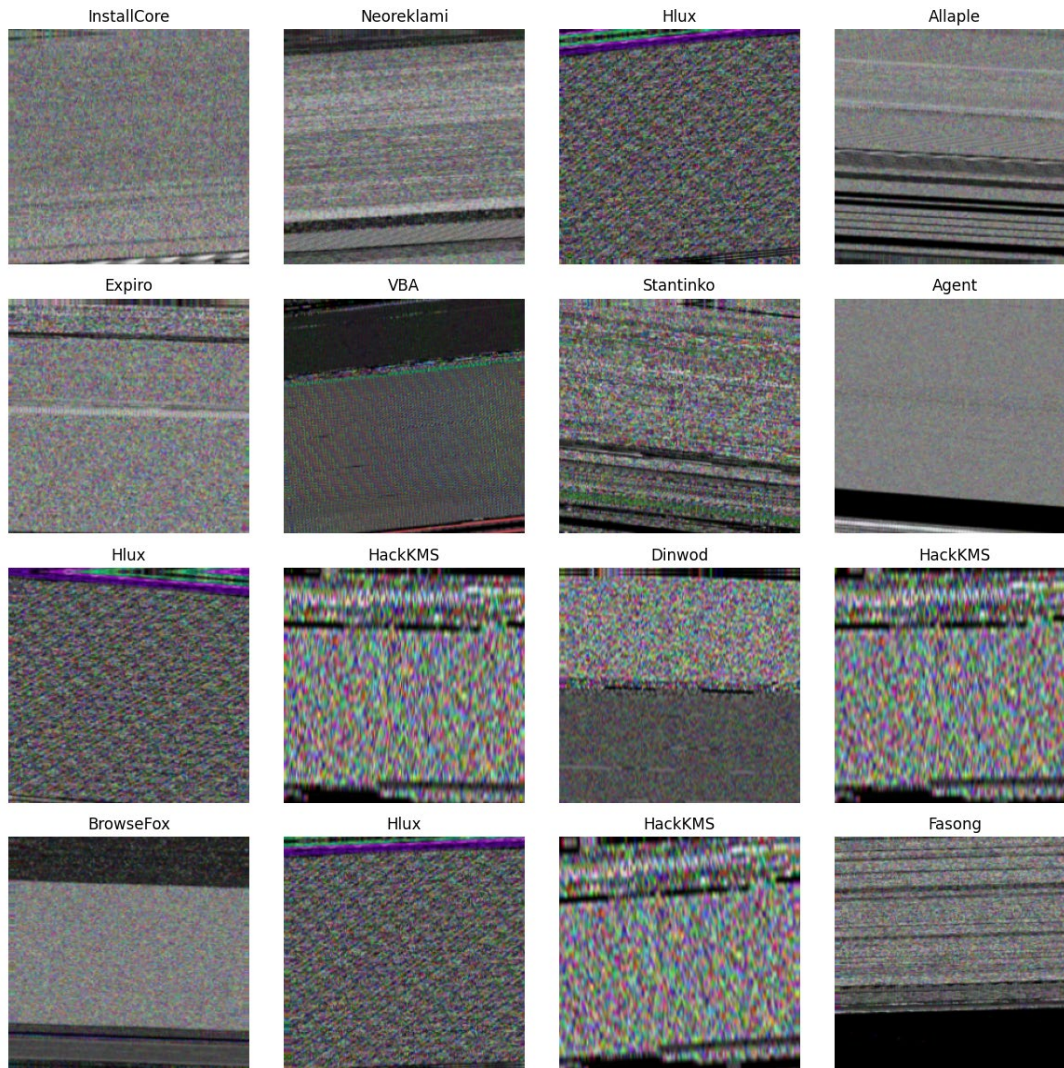


Figure 1: Representative MaleVis byte images. Horizontal bands, dark blocks, repeated textures

2.2 Duplicated Leakage and Corrected Split

The integrity audit identified 187 groups (480 unique images) of exact-duplicates. However, the most significant fact was that supplied training split contained exact matches for 173 validation images. By using the provided split without correction, we could reward memorization. You thus grouped exact-image hashes before you created the modeling partitions.

Chapter 3: Methodology

3.1 Input Pipeline

Images were resized to 224×224 and normalized to $[0,1]$. Training used batches of 32 with deterministic seeds. Development and test datasets were never augmented. The original training pipeline used mild random rotation (2%), zoom (3%), and translation (2%); whether these transformations were suitable for byte-image geometry was later tested directly.

3.2 Baselines(Efficeientnetb0 ,MobielNetV2)

Two lightweight transfer-learning models—EfficientNet-B0 and MobileNetV2—were used as additional baselines. Both models were fine-tuned on the same 25-class malware image dataset using $224 \times 224 \times 3$ inputs and a batch size of 32.

Among the two, EfficientNet-B0 performed better, achieving 85.68% test accuracy, compared to 81.25% for MobileNetV2. EfficientNet-B0 also had a slightly larger model size and parameter count, while MobileNetV2 remained the more compact baseline. These models provided a useful reference point for comparing the performance of the proposed CNN+CBAM approach.

3.3 CNN and CBAM architecture

The baseline is a four-block custom CNN. The blocks make use of both 32, 64, 128, and 256 3×3 filters, with batch norm, ReLU and max pooling. The global average pooling outputs a dense layer of 128 units that is followed by batch normalization, dropout and a softmax of 25 classes. After the fourth convolution, our attention model inserts CBAM. CBAM (Convolutional Block Attention Module) first gathers global average and maximum descriptors to estimate a channel coefficient, followed by estimating a spatial map from the resultant channel-refined features.

3.4 Optimization and Checkpoint

Adam was used as an initial learning rate of 5×10^{-4} , sparse categorical cross-entropy loss and loss-based checkpointing on validation-set. The first comparison of these settings was done with a 30 epoch limit the early-stopping patience was set to 6 and the learning-rate reduction factor was set to 0.5 after three epochs that did not hit a new low in model loss. Because of existing compute constraints, the subsequent ablation, tuning and cross validation run's employed a cap on epochs at 15 with early-stopping patience set to 4 in order to accomplish an experimental grid in time.

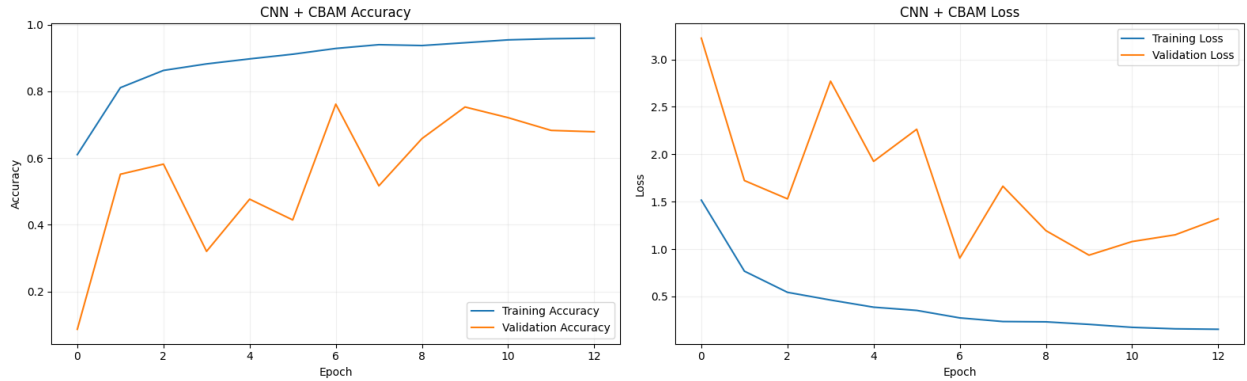


Figure 2 : CNN + CBAM learning curves. Validation volatility motivated checkpointing and restoration of the best validation-loss epoch

3.5 Evaluation Measures

Evaluations were performed with accuracy, balanced accuracy, macro and weighted precision/recall/F1, confusion matrices, one-vs-rest ROC-AUC, precision-recall curves log loss, Cohen's kappa and Matthews correlation coefficient. Macro-F1 was also focused on for comparing configurations because it weights all families equally. Final accuracy and macro-F1 were supplemented with 95% bootstrap confidence intervals from 1,000 resamples

Chapter 4: Experimental Design and Model Development

4.1 Initial controlled comparison

The first comparison modified a single architectural component: it included CBAM into the generalized CNN. The baseline trained for 21 epochs and restored epoch 15; the attention model

trained for 13 epochs and restored epoch 7. The frozen models were actually only used as references in the final study

4.2 Attention ablation

New architectures compare no attention to channel-only attention, spatial-only attention and full (CBAM). Key reference settings were held constant. In this run, full CBAM was best across all internal-validation metrics while the isolated channel (or spatial) module never exceeded the no-attention reference.

Configuration	Attention	Val. accuracy	Val. Macro-F1
Core CBAM	Channel + spatial	75.90%	74.92%
No attention	None	53.85%	52.00%
Channel only	Channel	53.11%	51.82%
Spatial only	Spatial	51.57%	44.52%

Table 3: Results of Attention Ablation

4.3 Controlled Tuning

CBAM reference was then adjusted one factor at a time: dropout 0.00, dropout 0.50, no batch normalization, five convolution blocks and off geometric augmentation. Without augmentation, the validation macro-F1 is a large margin higher. This final configuration kept 4 blocks and batch normalization

Experiment	Single Change	Val. accuracy	Val. Macro-F1
Augmentation off	Disable geometric transforms	92.75%	92.81%
Dropout off	0.00	76.30%	74.52%
Dropout 0.50	0.50	73.73%	72.30%
Five blocks	Add a 256-filter block	60.48%	56.29%
Batch norm off	Disable batch normalization	55.68%	54.55%

Table 4: Results of Controlled Tuning

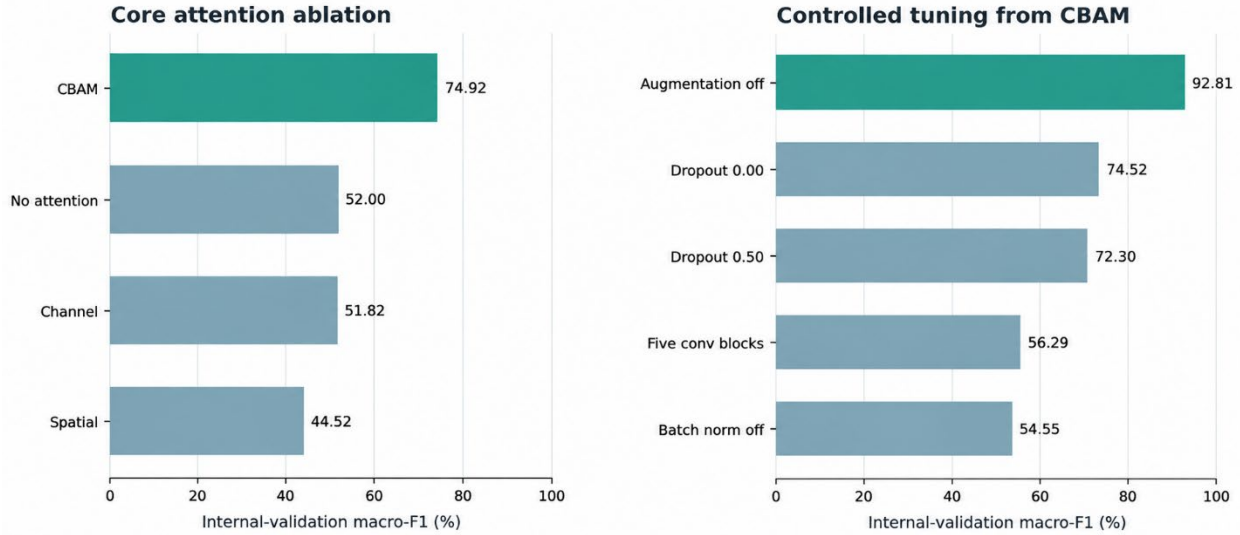


Figure 3: Comparing between Core attention ablation and Controlled tuning

4.4 Grouped five-fold validation and significance

The whole 8,750-images development pool was assessed using stratified Group K Fold with exact-image hash as group variable. Each fold was trained a corresponding no-attention model and the chosen CBAM model for ten runs. We matched and compared fold macro-F1 values using a two-sided Wilcoxon signed-rank test at $\alpha = 0.05$

Chapter 5: Result and Comparative Analysis

5.1 Initial Baseline and Attention Results

Metric	Baseline CNN	CNN+CBAM	Change
Test accuracy	77.25%	79.81%	+2.56 pp
Balanced accuracy	77.87%	77.40%	-0.47 pp
Macro precision	83.26%	81.76%	-1.50 pp
Macro recall	77.87%	77.40%	-0.47 pp
Macro-F1	77.30%	76.77%	-0.52 pp
Weighted-F1	76.56%	79.36%	+2.80 pp
Parameters	426,489	435,051	+8,562

Table 5: Results of CNN baseline and CNN+CBAM

5.2. Five-Fold Cross Validation

5 fold cross validation was run 10 times between No attention and Selected CBAM

Fold	No attention	Selected CBAM	Difference
1	92.19%	95.74%	+3.55 pp
2	90.62%	97.05%	+6.43 pp
3	94.70%	96.68%	+1.98 pp
4	92.21%	95.81%	+3.60 pp
5	95.56%	96.13%	+0.57 pp

Table 6: Results of 5 fold CV

5.3 Final held-out performance

After validation-only selection, it was the final augmentation-off CBAM that was evaluated as the best final model. For this, CBAM checkpoint evaluated on the 3,477-image test set. The model retained 435,051 parameters and required 10.82 seconds for inference, or approximately 3.11 ms per image

Metric	Final model value
Accuracy	92.69%
Balanced accuracy	92.87%
Macro precision / recall	94.25% / 92.87%
Macro-F1	93.00%
Weighted precision	94.03%
Weighted Recall	92.69%
Weighted F1	92.76%
Macro ROC-AUC (OvR)	0.9954
Cohen’s kappa / MCC	0.9238 / 0.9244

Table 7: Final Result

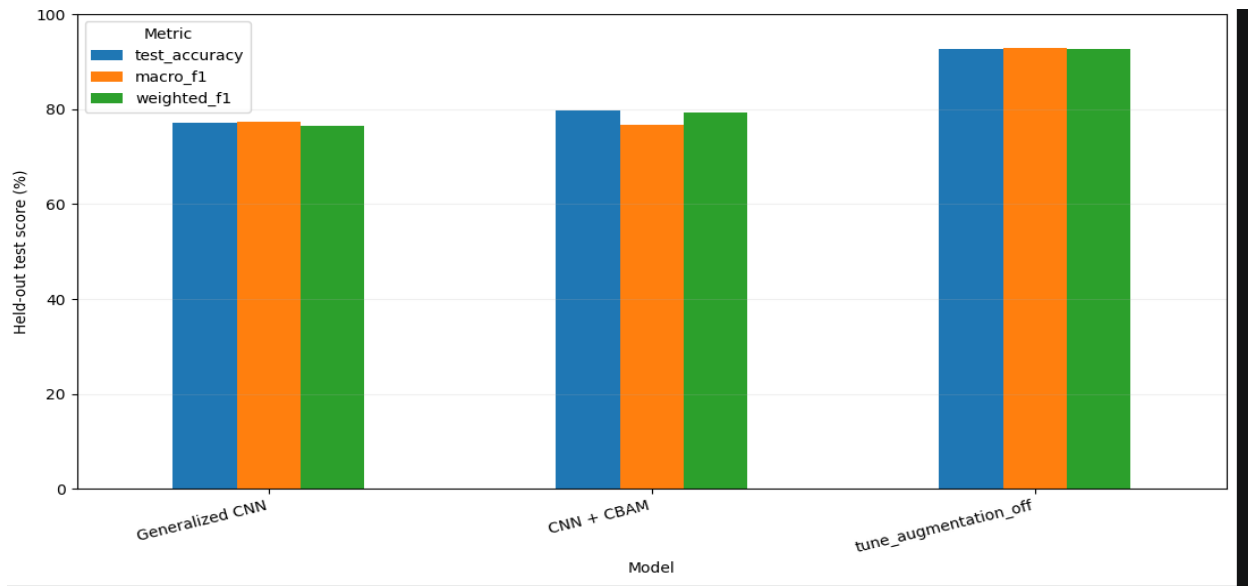


Figure 4: From baseline CNN and first attention model to the final selected model

Difference with final model	Accuracy	Macro-F1	Weighted F1
Versus baseline CNN	+15.44 pp	+15.70 pp	+16.21 pp
Versus initial CNN+CBAM	+12.88 pp	+16.22 pp	+13.40 pp

Table 8: Difference between baseline and CNN+CBAM with the final model

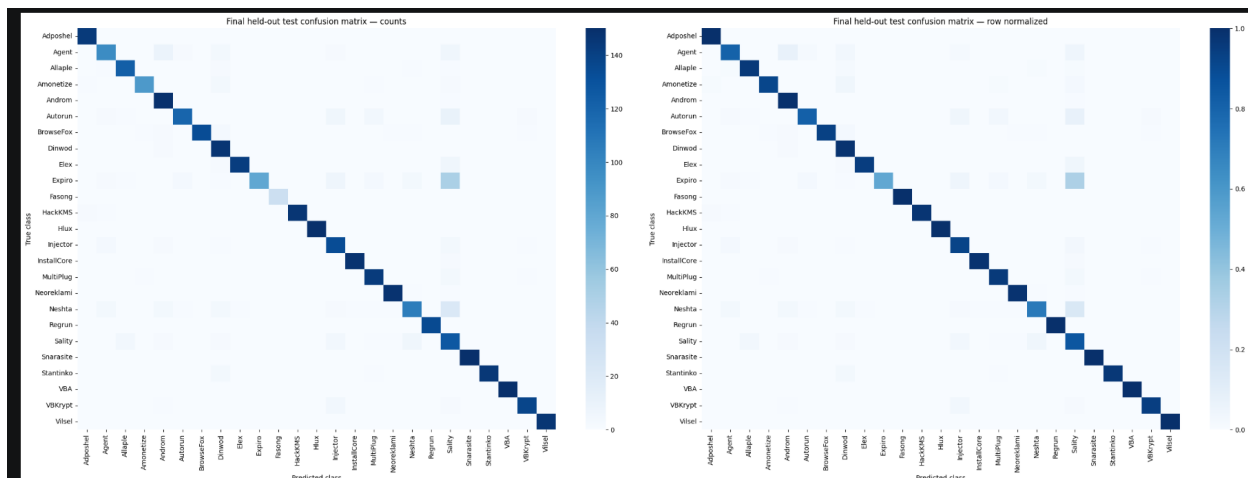


Figure 5. Final held-out confusion matrices in counts and row-normalized form

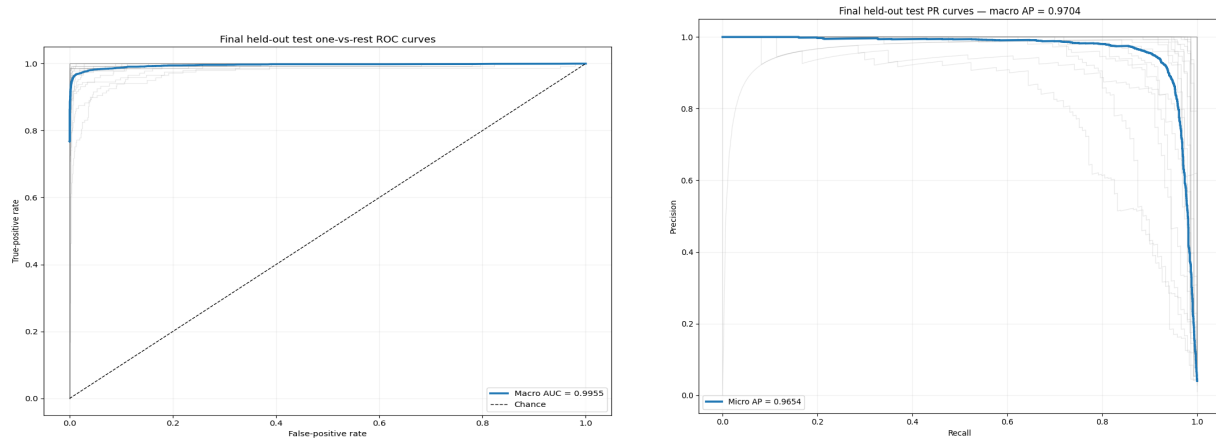


Figure 6: Final one-vs-rest ROC and precision–recall curves. Macro ROC-AUC is 0.9954 and macro average precision is approximately

5.4 Related Work Context

The five related malware-image studies, are reduced to summary form below. Four directly use MaleVis, and another one is supporting evidence for lightweight image preprocessing. Since the reported results differ in terms of datasets, splits, classes and model capacities, these are contextual according to their research designs.

Study	Main Method	Reported Result	Relation to this project
Ahmed et al. (2022) [10]	SFTA/Gabor texture features with GDA and Naive Bayes	About 98% multiclass accuracy with SFTA-GDA	Classical MaleVis baseline; motivates automated end-to-end CNN features

Atitallah et al. (2022) [11]	Three transfer-CNN backbones with random-forest voting	98.68% accuracy; 98.70% F1	Strong but resource-heavy ensemble using a different split
Liu et al. (2024) [12]	Compact CNN with weighted sampling and augmentation	83.22% MaleVis accuracy; 96.76% after removing the 'Other' class	Supports imbalance handling and class-level error analysis
Son et al. (2022) [13]	Low-dimensional images with k-NN, SVM, and a small CNN	98.51% on Maling with SVM; MaleVis not used	Indirect support for lightweight preprocessing and lower training cost
Alshomrani et al. (2025) [14]	ConvNeXt-Tiny and Swin Transformer with Grad-CAM	88.65% MaleVis accuracy; 92.69% F1	Closest explainable hybrid reference, but computationally heavier

Table 9: Related Work Table

Chapter 6: Explainability Analysis

6.1 Methods and selected cases

Grad-CAM produced a class-discriminative heat maps using the last convolutional feature layer, conv_block_4. LIME utilized 1,000 samples of the scene to approximately find super pixels that are locally supportive. Both methods were tested on a correctly predicted MultiPlug image and an incorrectly classified Neoreklami image estimated as Sality. These aren't even in color because analytical heatmap colors demonstrate the intensity of influence and have been retained.

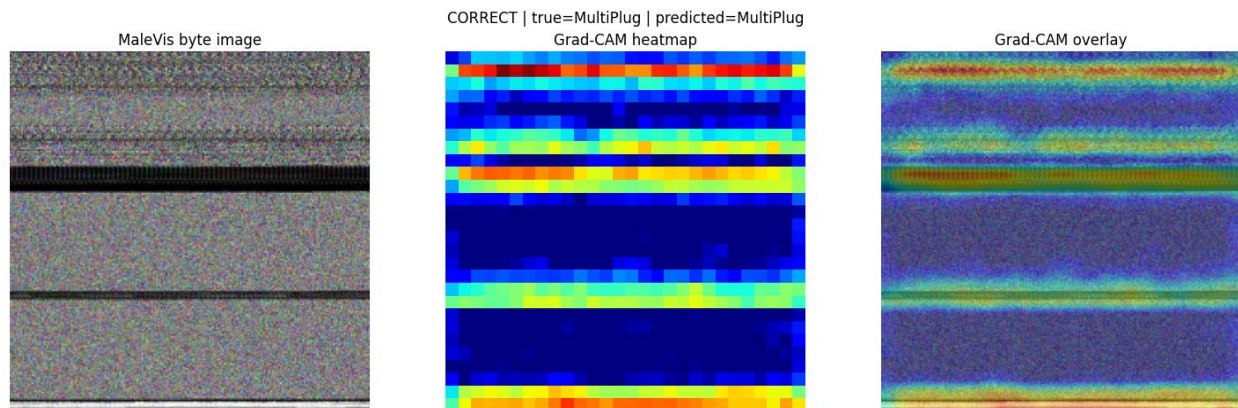


Figure 7: Correct MultiPlug prediction. Grad-CAM emphasizes prominent horizontal bands and boundary transitions

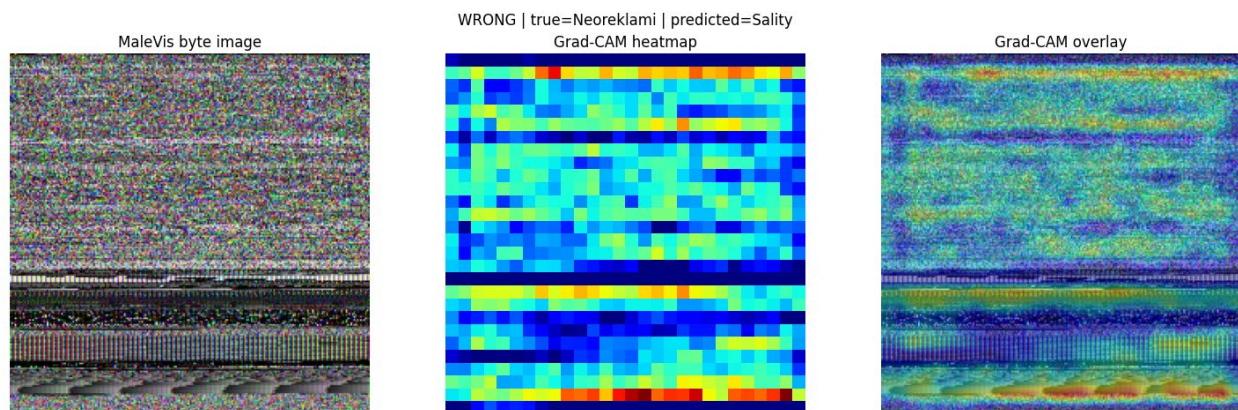


Figure 8: Incorrect Neoreklami→Salinity prediction. Activation is broader and includes strong lower-band structure [4].

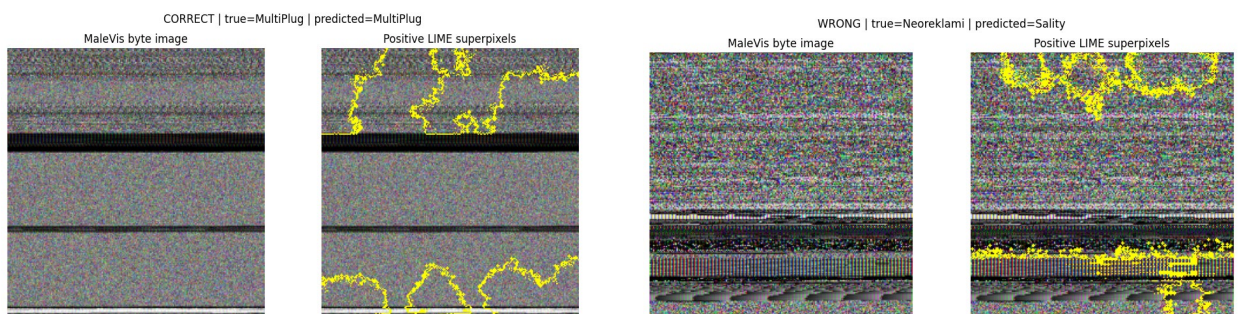


Figure 9: Positive LIME superpixels for the correct MultiPlug case (left) and incorrect Neoreklami→Salinity case (right)

In the case of a correct example, both Grad-CAM and LIME highlight structured bands and transitions. This qualitative match highlights that distributed byte-image structure, rather than just a single region in isolation, is required for prediction. Instead, in the wrong model, influential regions are more diffuse and overlap patterns that other models associate with Sality: That is evidence of local feature-space ambiguity.

Chapter 7: Discussion

7.1 Data integrity determined the validity

The duplicate audit was no simple data preprocessing detail. With-out hash grouping, repeated images could show up in training and evaluation alike making the notion of generalization meaningless. The final results are reasonable since the modeling partitions did not have exact-hash overlap and the test set was kept locked at selection time.

7.2 Attention helped, but the size of the claim depends on the comparison

The first CBAM by was able to improve the accuracy and weighted-F1 but not the macro-F1. We offer stronger evidence from our matched grouped-CV experiment later — attention improved macro-F1 by 3.23 percentage points on average across folds and decreased variability in every fold. Even so, $p = 0.0625$ does not permit a classic $\alpha = 0.05$ significance assertion. Conclusion: (PRACTICAL) This conclusion is that systematic improvement concurs with some evidence of significance at the fold level, but too few systems to achieve formal conclusions.

7.3 The Pipeline Effect

This great improvement of the final model over the previous models, is not entirely due to CBAM. It stands for model selection based on validation, well-matched augmentation choices, batch normalization kept, moderate dropout and architecture constraints. Disabling geometric augmentation was particularly crucial, presumably because operations that are label-preserving on natural images may break byte-layout structure. This explanation continues to be dataset dependent which must test externally.

7.4 Class-level evidence remains essential

The final macro-F1 and balanced accuracy are nearly equal to overall accuracy, which shows far greater balance compared to our previous models. Expiro and Sality, which you may have come across as important from the point of view of changing failure mode. As a result, you need to define class-specific false-positive and false-negative costs instead of one aggregate score for the production system.

7.5 Limitations

- Experiments are all performed with a single dataset and a single byte-to-image representation; performance across more recent malware collections or alternative renderings is not yet established.
- Exact-hash grouping naturally introduces no identical-file leakage but does not ensure that visually or semantically near-duplicate malware variants are grouped in a single partition as well.
- Five fold pairs have low power for significance tests and coarse exact p-values; repeat more grouped runs
- Compute time was explicitly limited in the tuning grid. Kernel sizes, optimizers, input sizes and wider learning-rate schedules were kept the same.
- Disabled Geometric Augmentation is contextual to the current render. Evaluate augmentations that are byte-aware and semantically-preserving.
- Numbers from related work are not directly comparable since the protocols and class sets differ.
- Grad-CAM and LIME cannot demonstrate causal malware behavior on their own: they are local approximations.
- Inference Time Measured in Kaggle but production calibration and drift, robustness and operational costs remain untested

Chapter 8: Conclusion

In this Project, an attention-enhanced CNN with leakage control mechanism was designed for the classification of 25-family MaleVis malware samples. The audit on the first data confirmed that it contained exact duplicates which rendered the provided split unsafe for a fair evaluation. An experimental framework grounded in hash grouping, a locked test set design, class-balanced metrics, and grouped cross-validation formed a more credible foundation.

Despite improving accuracy and weighted-F1, the initial CNN+CBAM comparison showed variable family-level gains. Fresh attention ablation selected full CBAM. Among the controlled one-factor tuning experiments, disabling geometric augmentation achieved the highest validation macro-F1. Finally, the model reached 92.69% held-out accuracy, 93.00% macro-F1 and 0.9954

macro ROC-AUC on the test set. In matched grouped cross-validation, CBAM increased mean macro-F1 over every single fold by 3.23 percentage points and was the winner from group F-4, yet the Wilcoxon p-value still remained at 0.0625 above significance threshold of 0.05.

Therefore, the final outcome supports this model selected as strongest in the configuration within this project but also keeping upon other valid statistical measures. Attribution methods such as Grad-CAM and LIME show that the predictions can be influenced by specific byte-image bands and transitions, but these explanations are local and non-causal. In summary, this project shows that class-agnostic model architecture is less crucial than sound data integrity, controlled comparison, and class-balanced evaluation —as well as reporting honest uncertainty.